

Supply Chain Finance and Resilience

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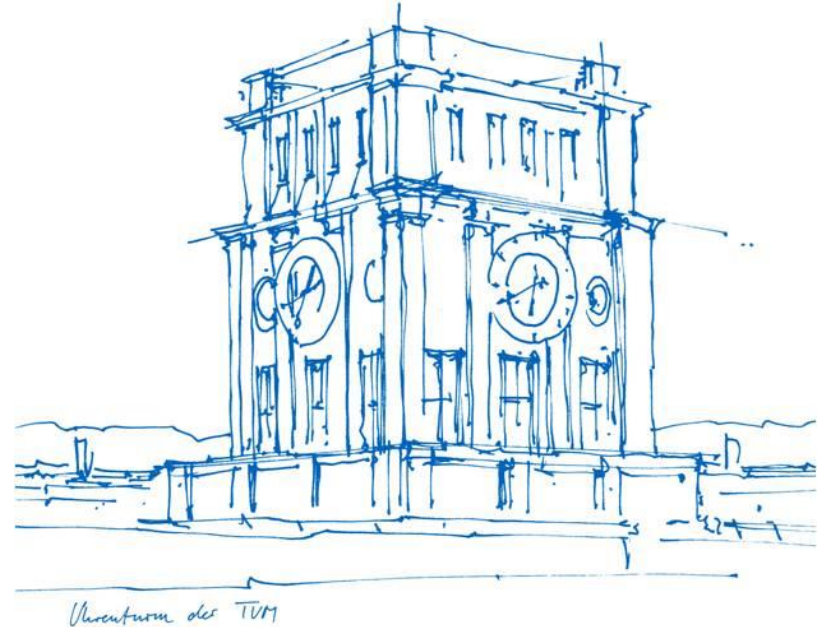
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Technische Universität München

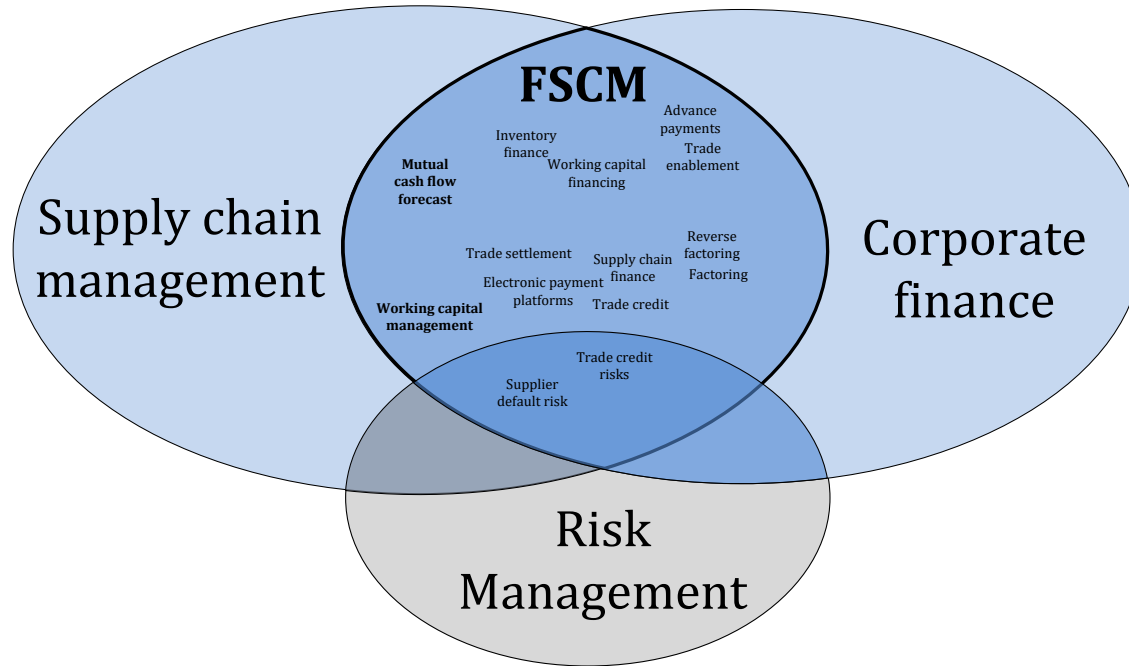
School of Management

Campus Heilbronn

Heilbronn, 2026



Concepts of Financial Supply Chain Management (FSCM)



What is Supply Chain Finance?

Difference between

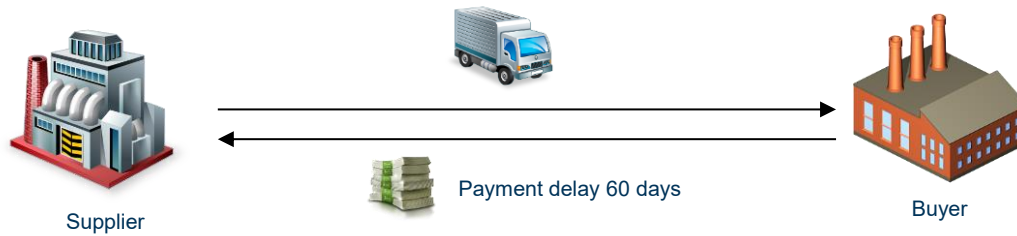
Supply Chain Finance

Supply Chain & Finance

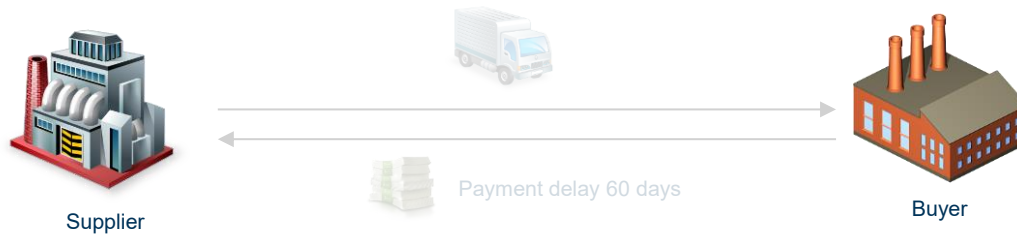
Supply Chain Financing

Motivation and Mechanism of Supply Chain Finance (Reverse Factoring)

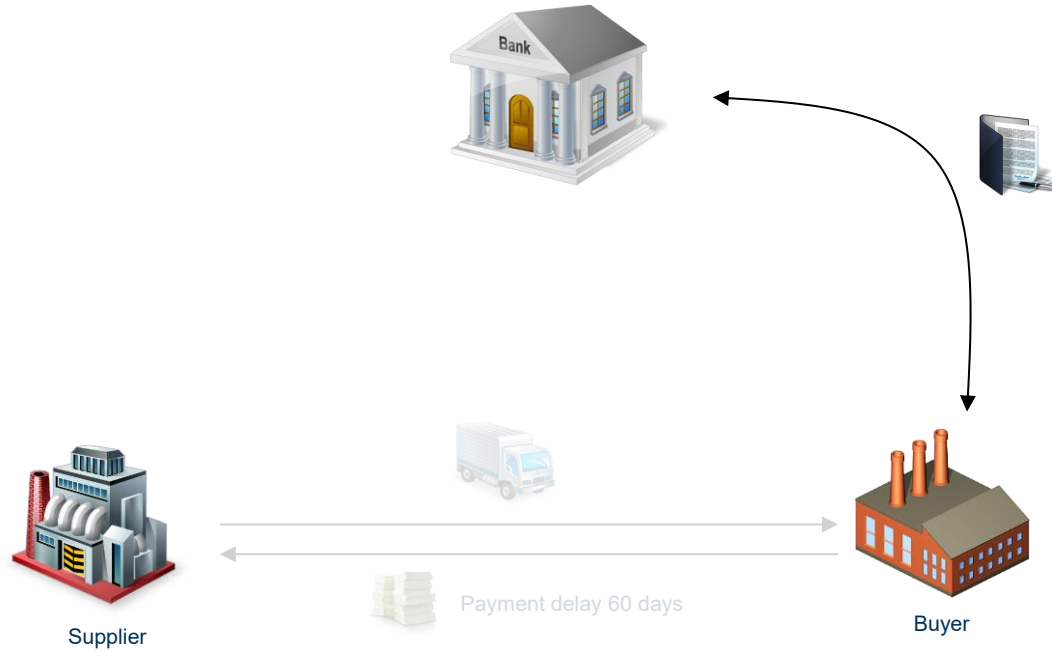
Supply Chain Finance



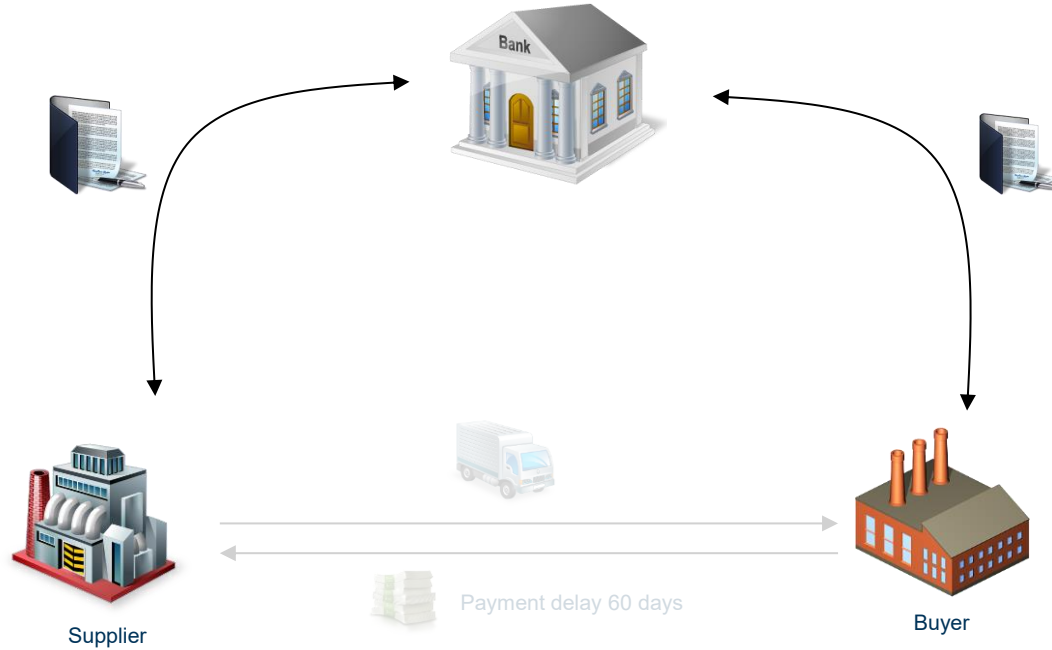
Supply Chain Finance



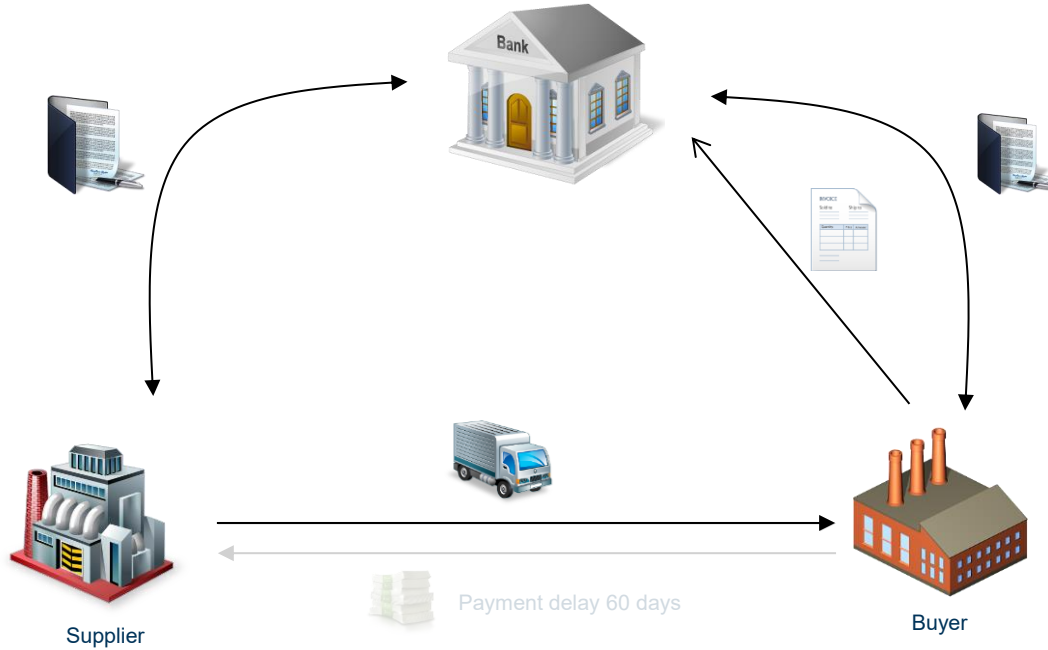
Supply Chain Finance



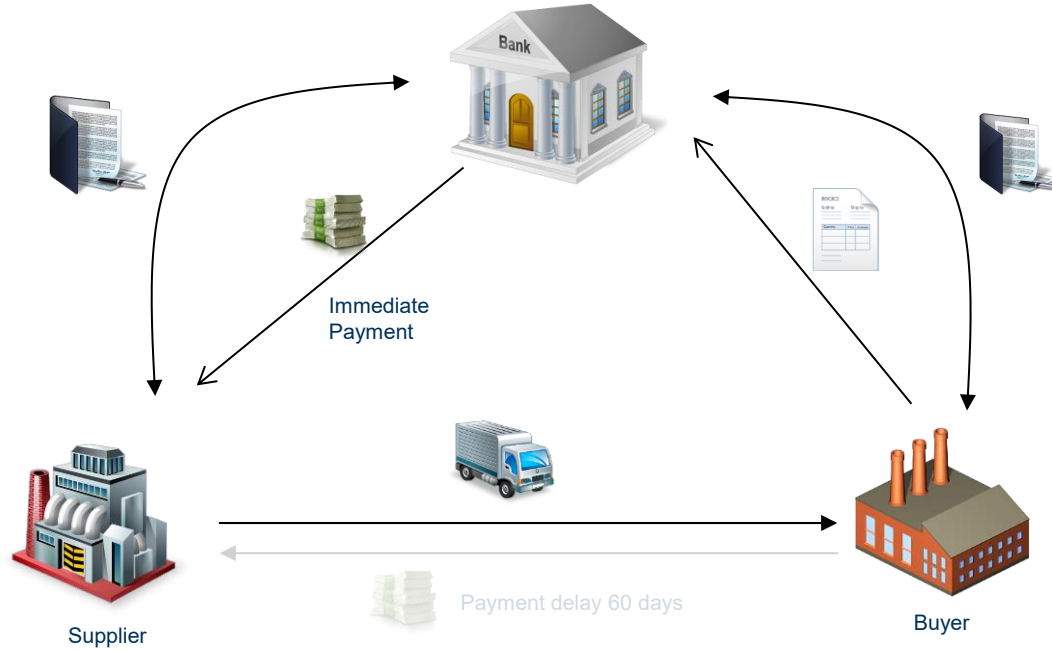
Supply Chain Finance



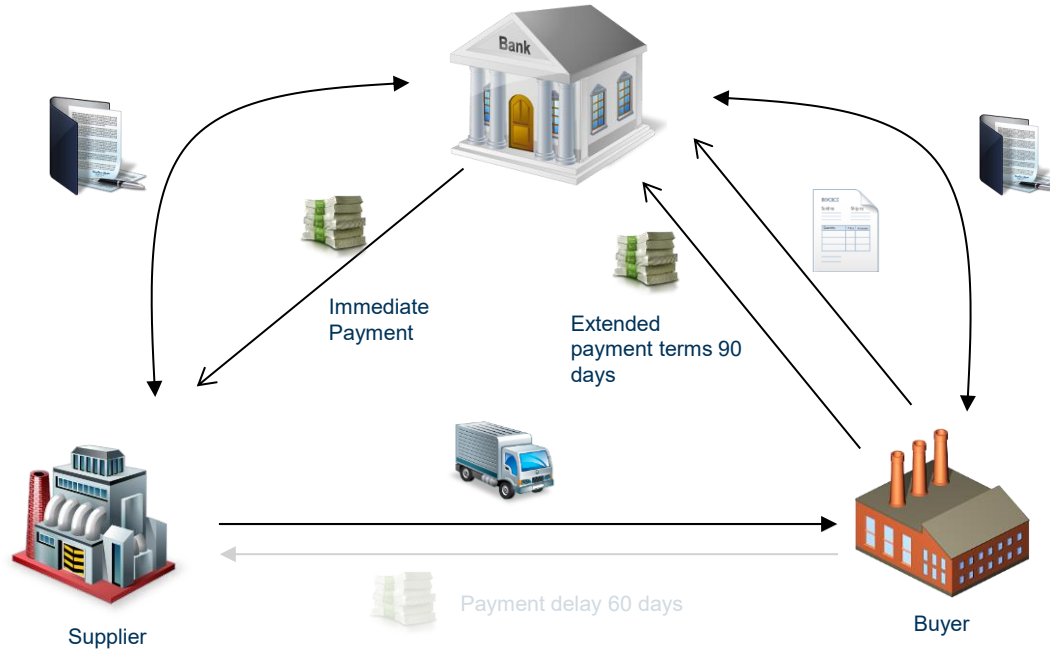
Supply Chain Finance



Supply Chain Finance



Supply Chain Finance



Supply Chain Finance Benefits (assume 500,000€ spend)

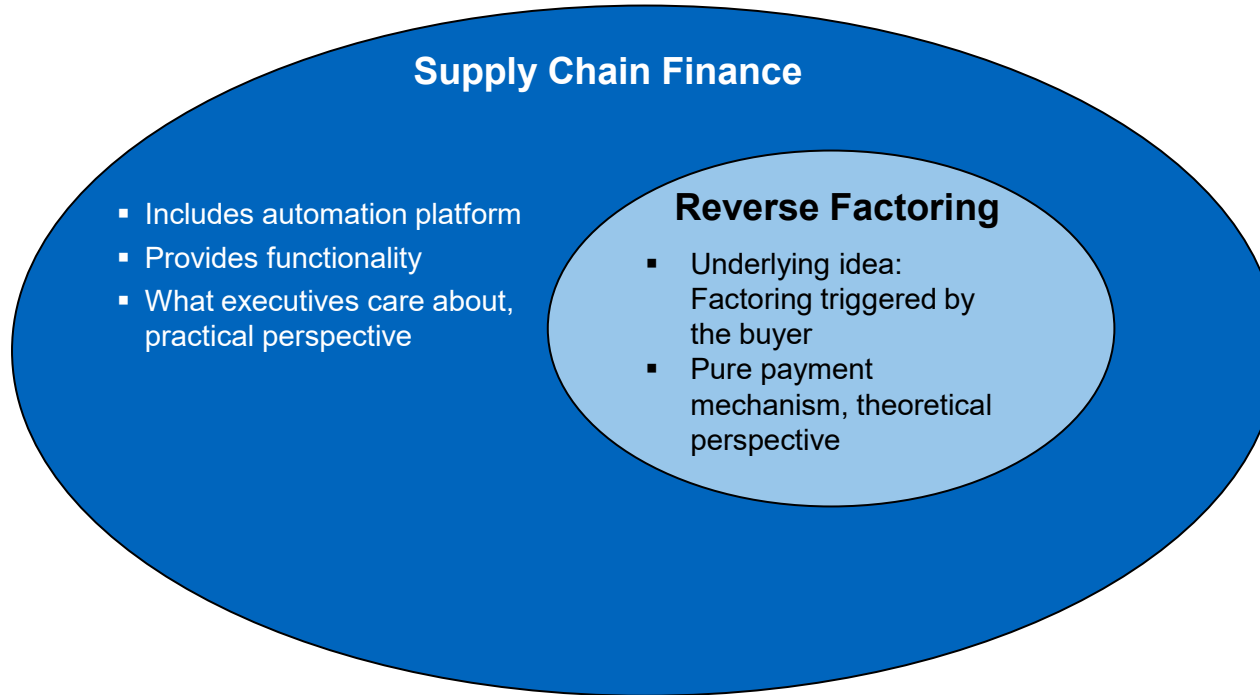
Without Supply Chain Finance

	Interest rate	Duration	Cost
Supplier	10%	60 days	€ 8,333
Buyer	4.5%	60 days	€ -3,750

With Supply Chain Finance

	Interest rate	Duration	Cost
Supplier	5%	90 days	€ 6,250
Buyer	4.5%	90 days	€ -5,625

Factoring – Reverse Factoring – Supply Chain Finance



An Empirical Analysis of Supply Chain Finance Adoption*

*) Based on Wuttke, D; Rosenzweig, E; Heese, S. (2019) An Empirical Analysis of Supply Chain Finance Adoption, *Journal of Operations Management* 65 (3), pp. 242-261

“

In fact, research shows that [...] buyers (those who set up an SCF program) consistently rank supplier onboarding as the single most important factor in a successful program.

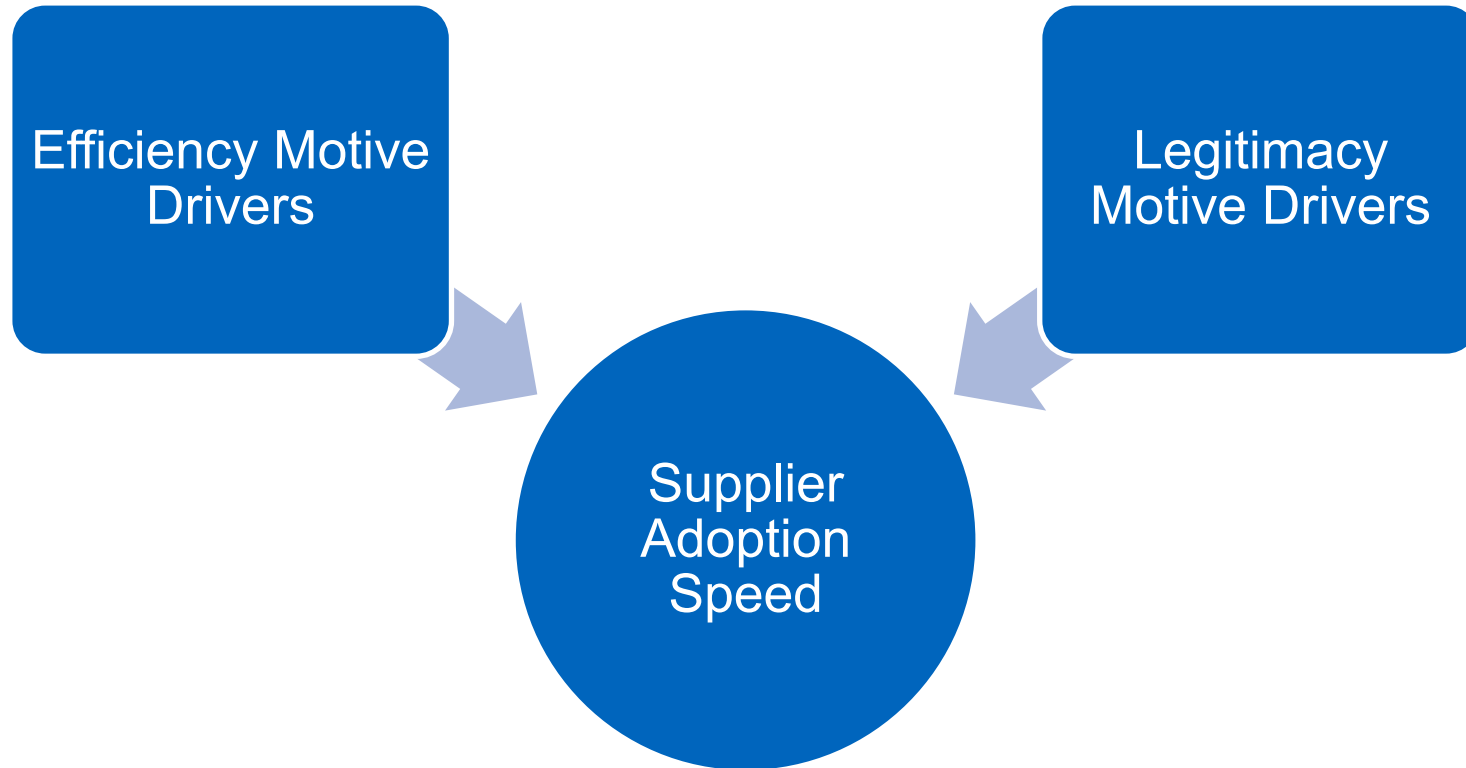
Herath (2015), McKinsey on Payments

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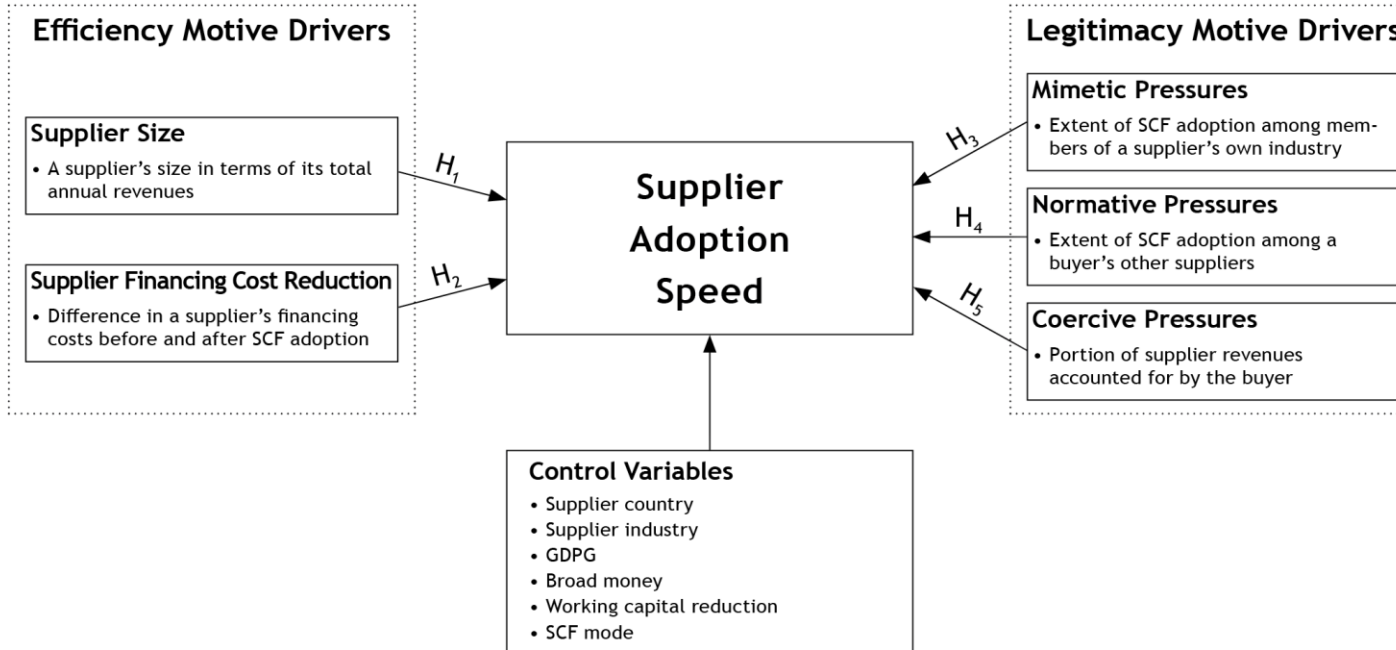
Research Questions

What are the key drivers of supplier adoption speed?

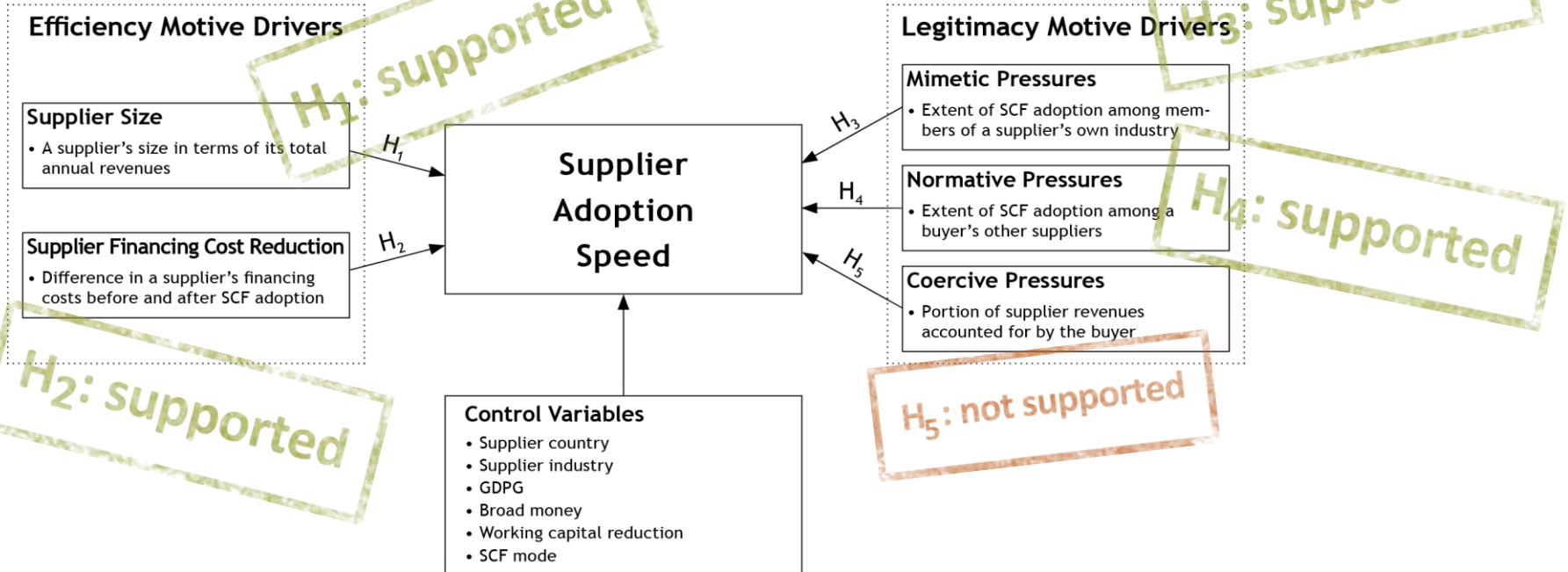
How can fast adopters be identified?



Conceptual Model



Conceptual Model



Managerial Implications

Buyers can achieve faster adoption

- Financing cost reductions are effective
- Smaller suppliers adopt faster: approach them first to build up normative pressures
- Mimetic pressures: target suppliers in industries with higher SCF adoption rates earlier

Buyers should avoid common pitfalls

- Do not underestimate the time it takes to get suppliers on board
- Do not go straight for the largest supplier
- Do not be focused too much on your own benefits at the beginning

Suppliers could

- Generate legitimacy
- Understand that substantial payment terms extensions are quite usual

SCF providers could

- Extend supplier-portfolio approach by including time dimension

SCF Criticism

The New York Times | <http://nyti.ms/1xYZyGf>

BUSINESS DAY

Big Companies Pay Later, Squeezing Their Suppliers

By **STEPHANIE STROM** APRIL 6, 2015

How would you like to have 120 days to pay your creditors?

Source: Strom, S. (2015). Big companies pay later, squeezing their suppliers. The New York Times, April 6.

Now, Suppliers Are Wal-Mart's Main Squeeze

- ▶ The company wants brands to pay up for space in its warehouses
- ▶ "It looks as though they are... trying to pad their own margins"

Supplying Wal-Mart Stores has never been easy. The \$486 billion retailer is so large that, for some brands, challenging

Source: Pettypiece, S., Townsend, M., & Cloeman-Lochner, L. (2015, 21.09.2015).
Now, Suppliers are Wal-Mart's Main Squeeze. *Businessweek*.

Supply Chain Resilience

PPR TWL WHT 2PLY 1
24 54 CT
0000-37000-47796
UNIT PRICE
\$4.61 ²⁵
2.49

LIMIT OF

1

MULTI-PACK

PAPER

TOWELS

LIMIT 2 SINGLE ROW

Supply chain glitches

Mean abnormal return of a supply chain glitch

–10.28 %, 1989 – 2000

(Hendricks and Singhal, 2003)

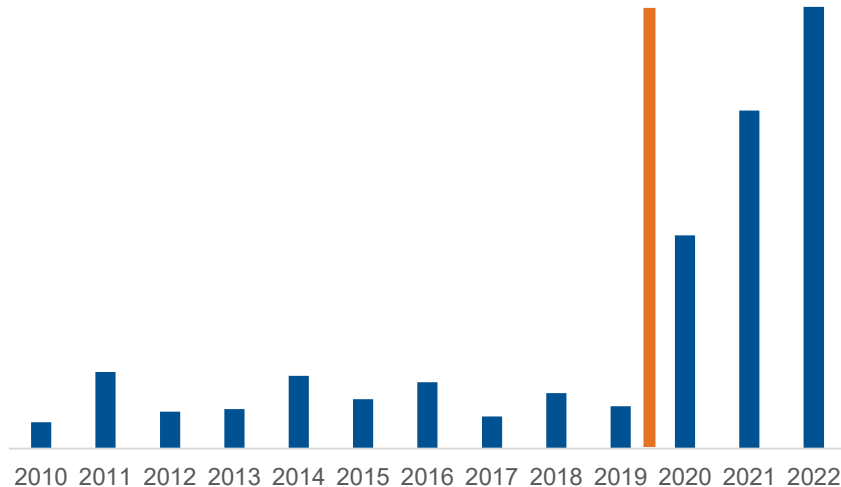
–3.55 %, 2013 – 2017

(Schmidt, Wuttke, Ball, and Heese, 2020)

Hendricks, K. B., & Singhal, V. R. (2003). The effect of supply chain glitches on shareholder wealth. *Journal of Operations Management*, 21(5), 501-522.

Schmidt, C. G., Wuttke, D. A., Ball, G. P., & Heese, H. S. (2020). Does social media elevate supply chain importance? An empirical examination of supply chain glitches, Twitter reactions, and stock market returns. *Journal of Operations Management*, 66(6), 646-669.

Supply chain resilience, Google search trend, worldwide



<https://trends.google.de/trends/explore?date=2010-01-01%202022-12-31&q=%22supply%20chain%20resilience%22>, as of 02/04/2024

Metaphorical resilience



Resilience definitions



“A firm’s ability to recover from disruptive events.”

(Blackhurst et al., 2011, p.374)

“...the capacity for an enterprise to survive, adapt, and grow in the face of change and uncertainty.”

(Petit et al., 2013, p. 70)

“...respond to unexpected disruptions and restore normal supply network operations.”

(Rice and Caniato, 2003, pp. 22-23)

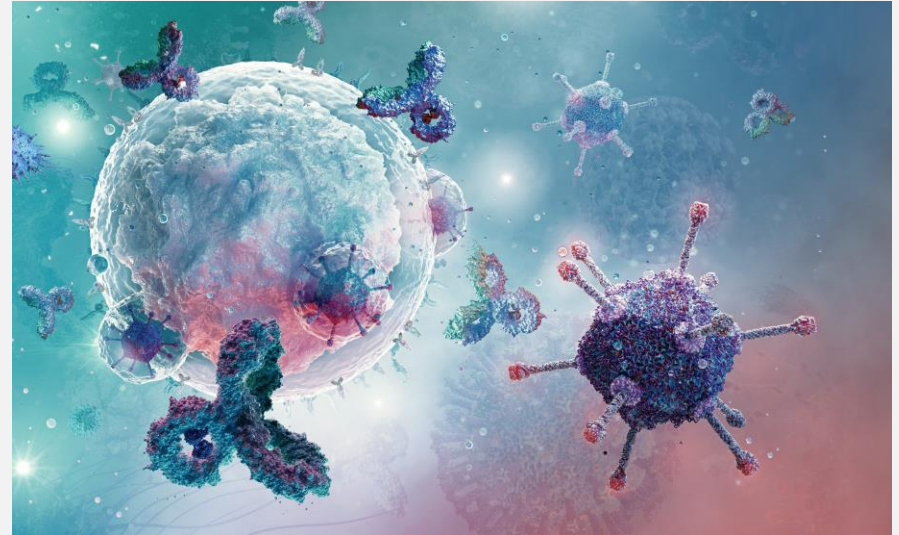
Blackhurst, J., Dunn, K. S., & Craighead, C. W. (2011). An empirically derived framework of global supply resiliency. *Journal of Business Logistics*, 32(4), 374-391.

Pettit, T. J., Croxton, K. L., & Fiksel, J. (2013). Ensuring supply chain resilience: development and implementation of an assessment tool. *Journal of Business Logistics*, 34(1), 46-76.

Rice, J. B., & Caniato, F. (2003). Building a secure and resilient supply network. *SUPPLY CHAIN MANAGEMENT REVIEW*, V. 7, NO. 5 (SEPT./OCT. 2003), P. 22-30.

A new resilience metaphor

“Resilience is the immune system of supply chains”



The impact of lean management on resilience

“Consider a lean supply chain. While hare-bones inventory levels decrease the impact of overforecasting demand, they simultaneously increase the impact of a supply chain disruption.”

(Chopra and Sodhi, 2004, p. 54)

“Extreme leanness and efficiency may result in increasing the level of vulnerability, at both the individual firm level and across the supply chain.”

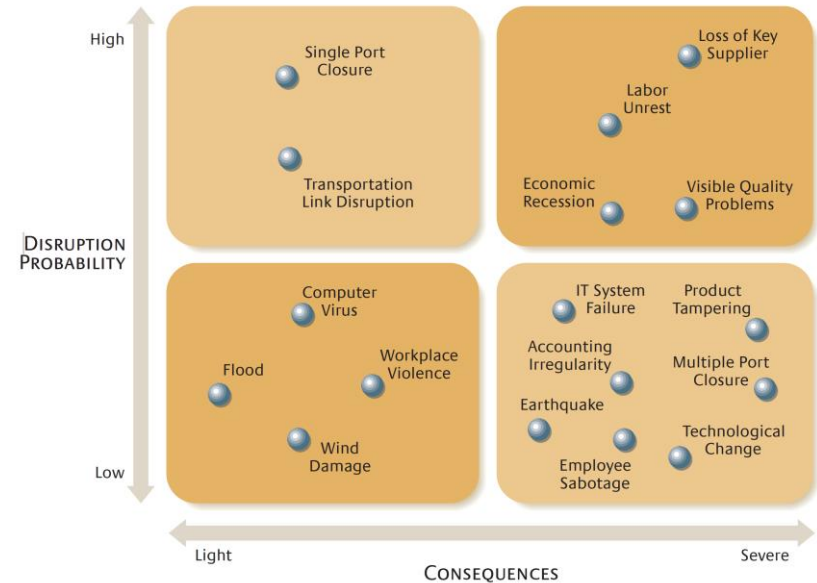
(Kleindorfer and Saad, 2005, p. 55)

Kleindorfer, P. R., & Saad, G. H. (2005). Managing disruption risks in supply chains. *Production and Operations Management*, 14(1), 53-68.
Chopra, S., & Sodhi, M. (2004). Supply-chain breakdown. *MIT Sloan management review*, 46(1), 53-61.

Sheffi and Rice (2005) vulnerability map

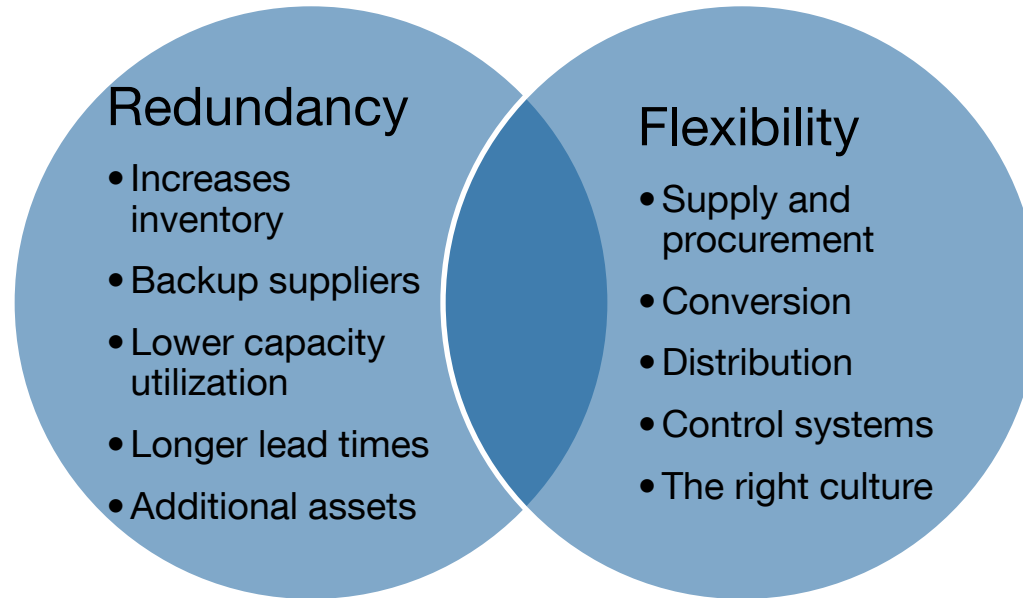
“An enterprise vulnerability map categorizes the relative likelihood of potential threats to an organization and the company’s relative resilience to such disruptions. Such maps can then direct management attention and prioritize the planning.”

(Sheffi and Rice, 2005, p. 44)



Sheffi, Y., & Rice Jr, J. B. (2005). A supply chain view of the resilient enterprise. *MIT Sloan management review*, 47(1), 41.

Two strategies towards resilience



Sheffi, Y., & Rice Jr, J. B. (2005). A supply chain view of the resilient enterprise. *MIT Sloan management review*, 47(1), 41.

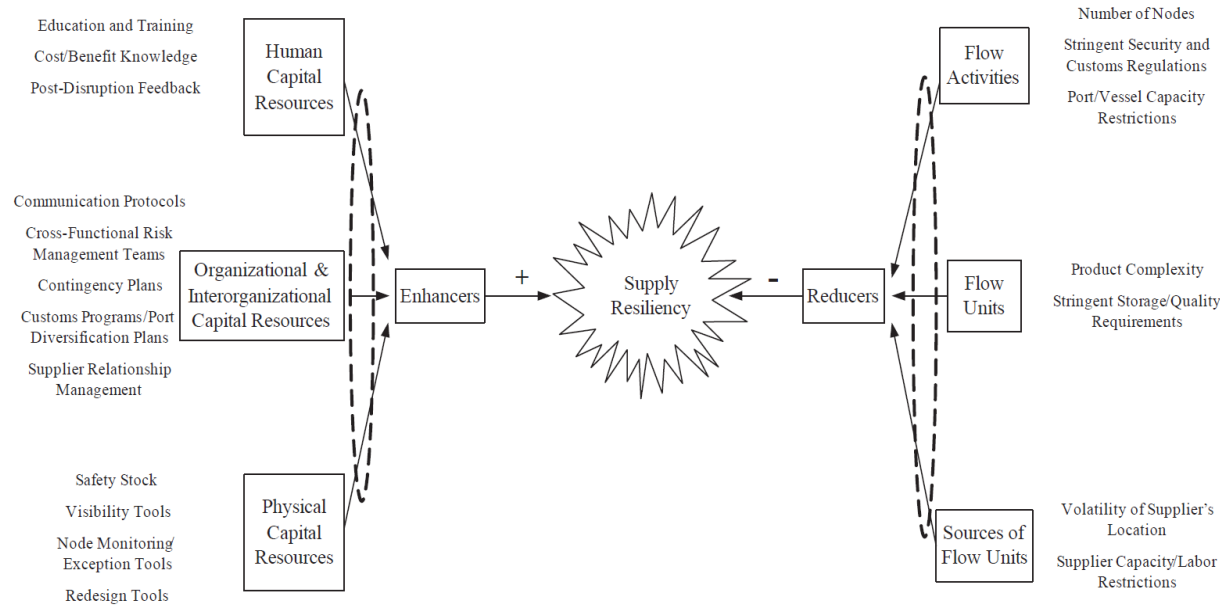
Can firms achieve supply chain resilience efficiently?

“The fifth principle implies that in order to minimize risk, and ultimately loss from supply chain disruptions, attention must be given to the tradeoff between ‘**robustness**’ of the supply chain to disruptions and the overall **efficiency** of the supply chain under normal operations.”

(Kleindorfer and Saad, 2005)

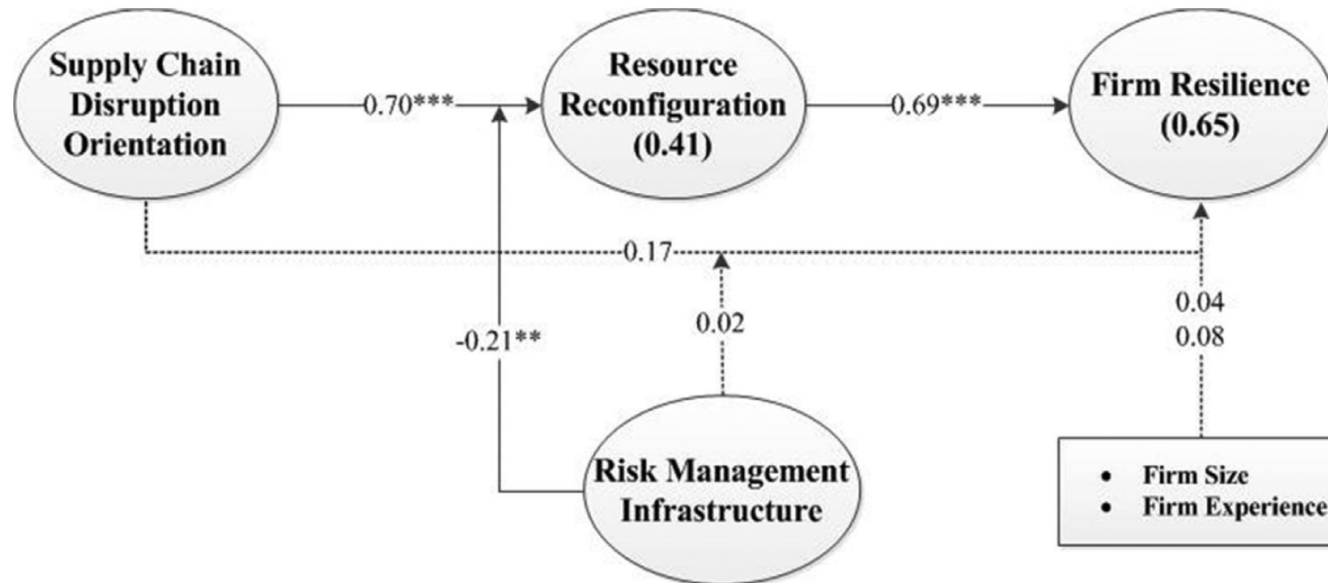
Kleindorfer, P. R., & Saad, G. H. (2005). Managing disruption risks in supply chains. *Production and Operations Management*, 14(1), 53-68.

Blackhurst et al. (2011) framework of supply chain resilience



Blackhurst, J., Dunn, K. S., & Craighead, C. W. (2011). An empirically derived framework of global supply resiliency. *Journal of Business Logistics*, 32(4), 374-391.

Ambulkar et al. (2015) model of high impact disruptions



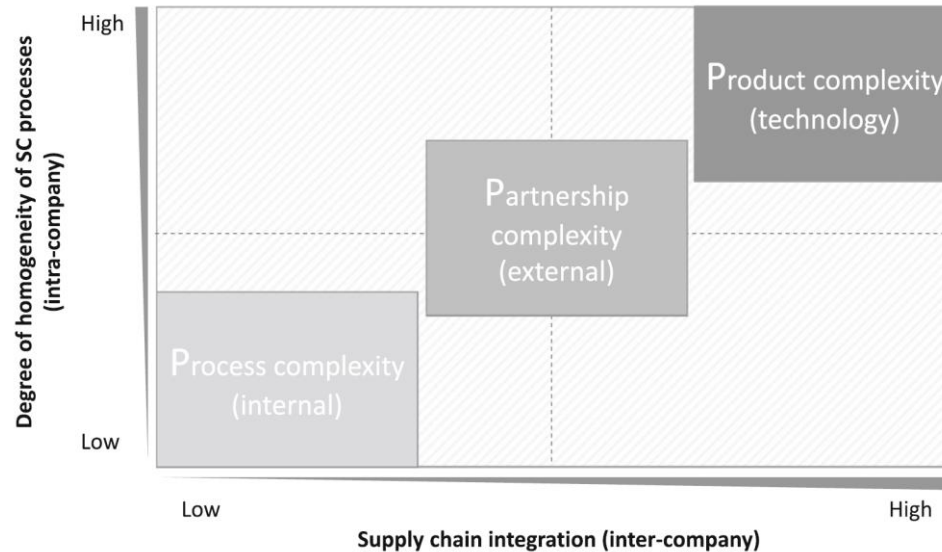
Ambulkar, S., Blackhurst, J., & Grawe, S. (2015). Firm's resilience to supply chain disruptions: Scale development and empirical examination. *Journal of Operations Management*, 33, 111-122.

The traditional paradigm – pick a solution

Supply Chain Resilience Responses by Failure Mode			
Resilience to Disruption in...	Action	Advantages	Disadvantages
Supply	Use multiple and/or local sources in different locations.	Spreads risk across two firms, two locations; local source protects against international supply shortages.	Higher cost to qualify supplier, lower volume leverage, no assurance additional supplier is more resilient.
	Use single source.	Known supplier, high supplier commitment, leveraged volume.	Vulnerable to disruption unless supplier has multiple flexible sites, backup plans.
	Contract for supplier flexibility.	Contract obligates supplier in advance.	Potentially higher cost per unit, may entail fixed costs for "take or pay" contracted volume. ⁶
	Modify inventory levels.	Right parts inventory and risk pooling may reduce inventory costs.	Requires periodic analysis by item as conditions change.
Transportation	Modify product to use standard parts.	Reduces part and inventory cost, complexity.	Costly to modify existing materials standards.
	Prepare for and use multiple modes and carriers.	Pre-disruption relationship ensures support in crisis.	May need to commit volume to the alternate mode to get access in a disruption.
	Use spot market for capacity.	Efficient transaction with no upfront or loading commitment.	Unknown carrier means added risk, potential for exceptional high pricing.
	Use logistics providers to source transportation.	Providers may have greater leverage and access.	Requires commitment (volume, cost) and relationship with logistics provider.
Production Facilities	Use multiple sites, each making multiple products.	Enables shifting production around locations.	Requires standardization in production operations, additional capital for additional facilities. ⁷
	Modify inventory levels and policies.	Right finished-goods inventory levels and risk pooling may reduce inventory costs.	Requires periodic analysis, potential redesign of supply network.
	Modify product to use standard processes.	Leverages common processing capabilities for lower cost, easier backup available.	Costly to modify product and production processes.
	Identify and contract backup production facilities.	Committed backup assured, potential to co-locate at supplier or customer.	Not dependable without contingency contract for the facilities in disruption. ⁸
Communications	Use range of communication media. ⁹	Communication in nearly any event.	Must maintain broad range of old and new technology.
	Back-up data.	Protects against data loss.	Still requires physical system in event of system loss.
Human Resources	Contract for backup IT system.	Provides for near-term system availability.	Potential delay in immediate response to massive system disruption.
	Set up and operate parallel or mirrored IT system.	Provides for immediate system availability.	Requires cost to build, operate, and maintain separate system in protected environment.
Human Resources	Develop cross-trained workers.	Enables shifting of employees and production as needed.	Must cross-train employees, and modify work system to utilize multi-skilled employees.
	Modify production process for unskilled labor.	Allows rapid increase or decrease in capacity.	Requires simplification of production process (not always feasible).
Human Resources	Back up knowledge.	Best practices captured and documented.	Requires significant investment to capture and maintain knowledge in useful form.

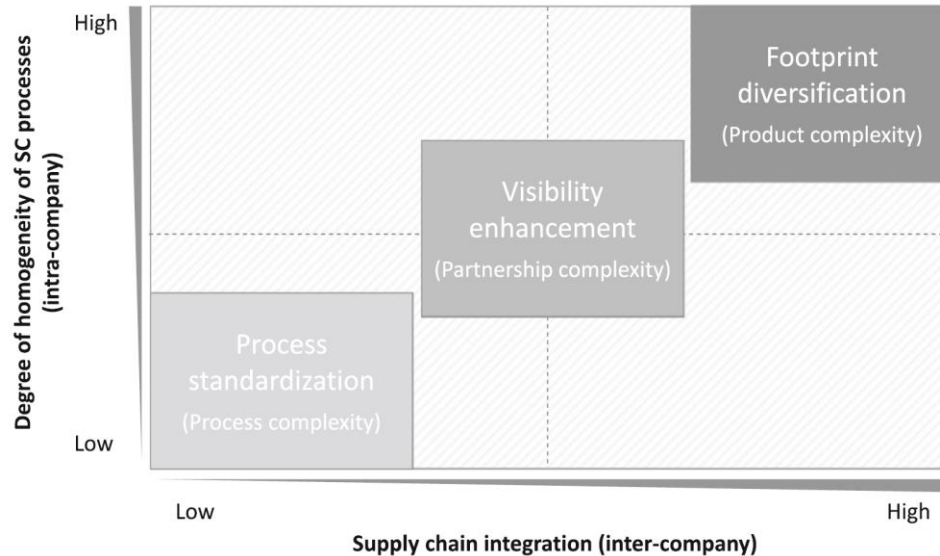
Rice, J. B., & Caniato, F. (2003). Building a secure and resilient supply network. *SUPPLY CHAIN MANAGEMENT REVIEW*, V. 7, NO. 5 (SEPT./OCT. 2003), P. 22-30: ILL.

Triple-p supply chain archetypes



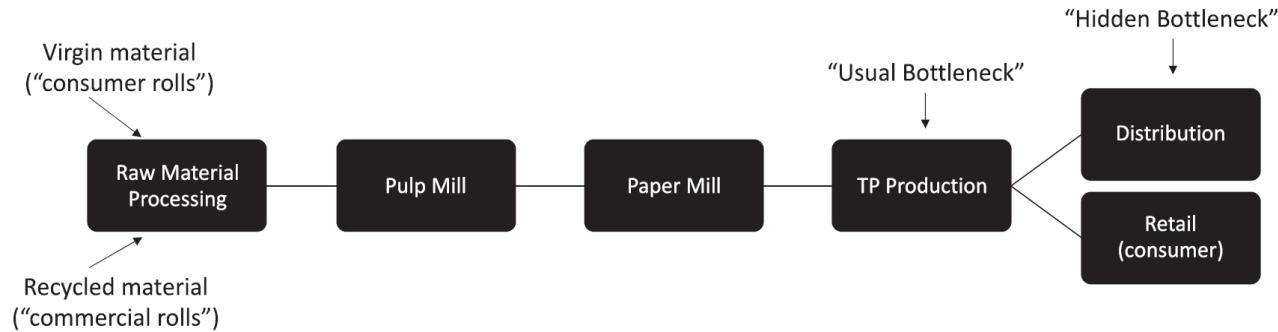
Cohen, M., Cui, S., Doetsch, S., Ernst, R., Huchzermeier, A., Kouvelis, P., . . . , Tsay, A. A. (2022). Bespoke supply-chain resilience: The gap between theory and practice. *Journal of Operations Management*, 68(5), 515-531.

Common resilience strategies based on triple-p archetypes



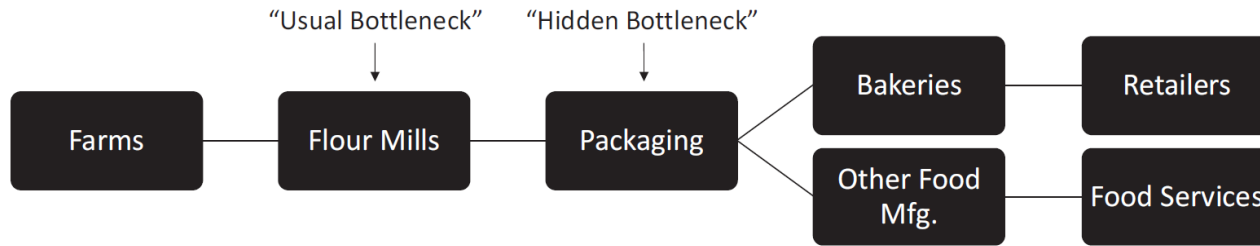
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Kouvelis' hidden bottlenecks: supply chain of toilet paper



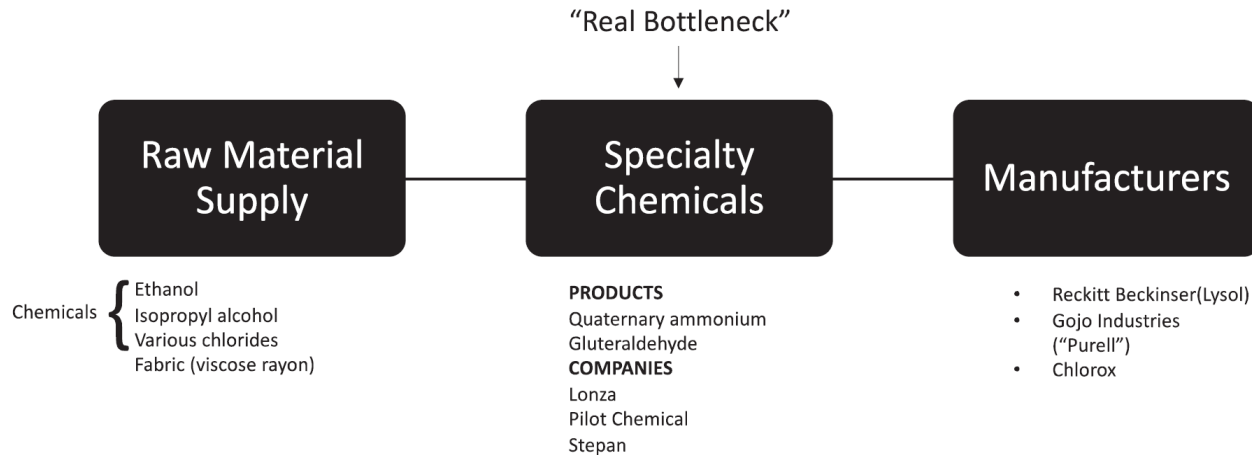
Kouvelis, P. (2022). Paradoxes and mysteries in virus-infected supply chains: Hidden bottlenecks, changing consumer behaviors, and other non-usual suspects. *Business Horizons*, 65(4), 469-479.

Kouvelis' hidden bottlenecks: supply chain of flour



Kouvelis, P. (2022). Paradoxes and mysteries in virus-infected supply chains: Hidden bottlenecks, changing consumer behaviors, and other non-usual suspects. *Business Horizons*, 65(4), 469-479.

Kouvelis' hidden bottlenecks: supply chain of disinfectants



Kouvelis, P. (2022). Paradoxes and mysteries in virus-infected supply chains: Hidden bottlenecks, changing consumer behaviors, and other non-usual suspects. *Business Horizons*, 65(4), 469-479.

References

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